Types of Mixtures

LAB LESSON

OBJECTIVE

With the help of an experiment, students will learn how to differentiate between homogeneous and heterogeneous mixtures.

ACARA CONTENT DESCRIPTIONS

Mixtures, including solutions, contain a combination of pure substances that can be separated using a range of techniques (ACSSU113)

- recognising the differences between pure substances and mixtures and identifying examples of each
- identifying the solvent and solute in solutions

Planning and Conducting:

Collaboratively and individually plan and conduct a range of investigation types, including fieldwork and experiments, ensuring safety and ethical guidelines are followed (ACSIS125)

- working collaboratively to decide how to approach an investigation
- learning and applying specific skills and rules relating to the safe use of scientific equipment
- identifying whether the use of their own observations and experiments or the use of other research materials is appropriate for their investigation

LESSON PLAN

Activities	Resources
Activity 1: How Is Matter Classified? Open the presentation to the second slide and review concepts taught in the previous lesson by asking students the following question: • How is matter classified? (Pure substances and mixtures) Use the question found on slide 3 to assess students' understanding of mixtures. 5	• Presentation: Types of Mixtures
Activity 2: Let's Make Some Mixtures Give out the Let's Make Some Mixtures 1 and 2 worksheets and divide students into groups. Briefly go through the instructions before allowing them to work through the investigations. 30	• Photocopies of the Let's Make Some Mixtures 1 and 2 worksheets • Sugar, iron filings, carbon dioxide, oil, water, sand, flour • 3 test tubes, straw, stirrer, paper towel, teaspoon
Activity 3: Is It Homogeneous or Heterogeneous? Before reviewing the answers for Activity 1, give out the All about Mixtures worksheet and play the video. The video will introduce students to the two different types of mixtures: homogeneous and heterogeneous. Ask them complete the worksheet after watching the video. Review all answers as a class with slide 5 of the presentation. 25	• Photocopies of the All about Mixtures worksheet • ClickView video *Mixtures* **https://clickv.ie./w/lp/7**



ANSWERS

Let's Make Some Mixtures 2

Test	What kind of mixture is this?	Can you see the individual components of each substance?	♠
A	solid-solid mixture	yes	heterogeneous
B	solid-liquid mixture	no	homogeneous
C	gas-liquid mixture	no	homogeneous
D	solid-liquid mixture	yes	heterogeneous
E	solid-liquid mixture	yes	heterogeneous
F	solid-liquid mixture	yes	heterogeneous
G	liquid-liquid mixture	no	heterogeneous

All about Mixtures

1. A homogeneous mixture is a mixture that does not allow you to see the individual particles easily as it has a uniform composition.

   A heterogeneous mixture is a mixture that is composed of particles that are not uniform in composition and can be seen easily.
2. Refer to table above.
3. a) 4
   b) 2
   c) 1
   d) 1
   e) 4
   f) 1
   g) 5
   h) 2
4. SOLUTION
5. Possible Answer:

Homogeneous mixture	apple juice, coffee, gravy
Heterogeneous mixture	cereal with milk, trail mix, muesli, salad, oil in water

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Let's Make Some Mixtures 1

Materials required:

Substances:

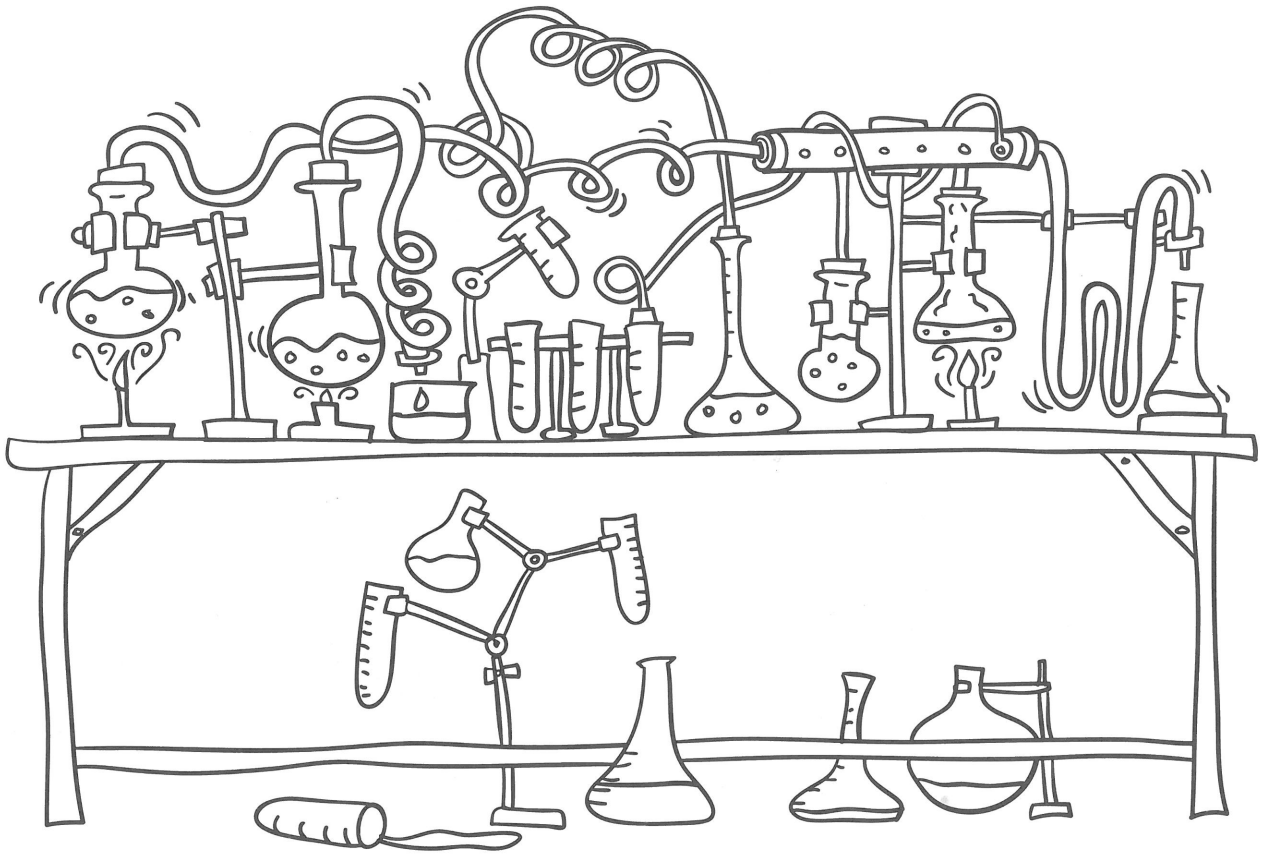
- sugar
- iron filings
- carbon dioxide
- oil (any type)
- tap water
- sand
- flour

Equipment:

- 3 x 15 mL test tubes
- straw (for blowing carbon dioxide)
- stirrer
- paper towel
- teaspoon

Method:

1. Complete Test A by mixing half a teaspoon of sugar with half a teaspoon of iron filings in a test tube. Use the stirrer to combine well.
2. Record your observations by completing the table found on the Let's Make Mixtures 2 worksheet.
3. Empty the contents of the test tube into the sink before rinsing and drying it with the paper towel.
4. Complete Tests B-G using the same method described above.

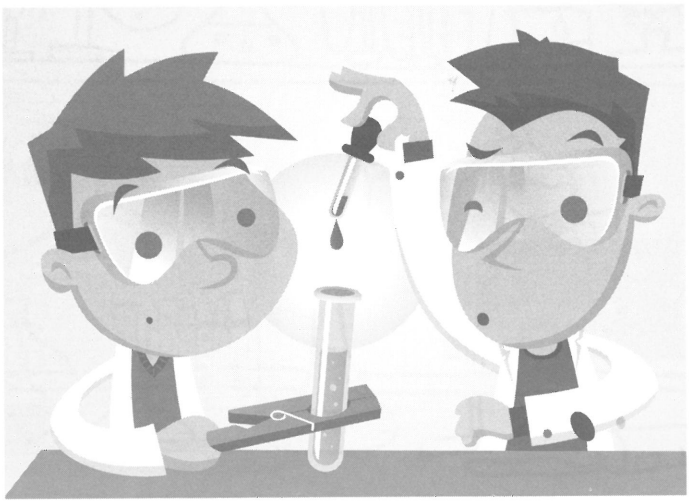


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Let's Make Some Mixtures 2

Complete Tests A-H by mixing substance 1 with substance 2 in a test tube. Fill in the table with your observations and results.

Test	Substance 1	Substance 2	What kind of mixture is this? (solid-solid, solid-liquid, etc.)	Can you see the individual components of each substance?	♠
A	½ tsp. sugar	½ tsp. iron filings			
B	½ tsp. sugar	10 mL tap water			
C	Carbon dioxide	10 mL tap water			
D	½ tsp. iron filings	10 mL tap water			
E	½ tsp. sand	10 mL tap water			
F	½ tsp. flour	10 mL tap water			
G	2 mL oil	10 mL tap water			



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All about Mixtures

Questions and Discussion

1. Complete the sentences using your own words.

A **homogeneous** mixture is a mixture that _____

_____

A **heterogeneous** mixture is a mixture that _____

_____

2. In the last column of the table (♠) in the Let's Make Some Mixtures 2 worksheet, state whether the mixture is a heterogeneous or homogeneous mixture.

3. By referring to your results from the experiment, indicate the correct answer to each question by shading the correct box. The first question has been answered for you.

a) How many different solid substances were used in total?	5 B	4 O
b) How many different liquid substances were used in total?	2 L	1 B
c) How many different gas substances were used in total?	0 C	1 S
d) How many solid-solid mixtures were formed?	1 U	3 Z
e) How many solid-liquid mixtures were formed?	3 R	4 O
f) How many liquid-liquid mixtures were formed?	1 T	2 P
g) How many heterogeneous mixtures were formed?	6 J	5 I
h) How many homogeneous mixtures were formed?	2 N	3 G

4. Use the letters found next to your answers above to complete the sentence below. Hint: the letters are not in the order they will be used.

A HOMOGENEOUS MIXTURE IS ALSO KNOWN AS A:

5. Write two examples of homogeneous and heterogeneous mixtures you see in your daily lives.

Homogeneous mixture	
Heterogeneous mixture	

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